

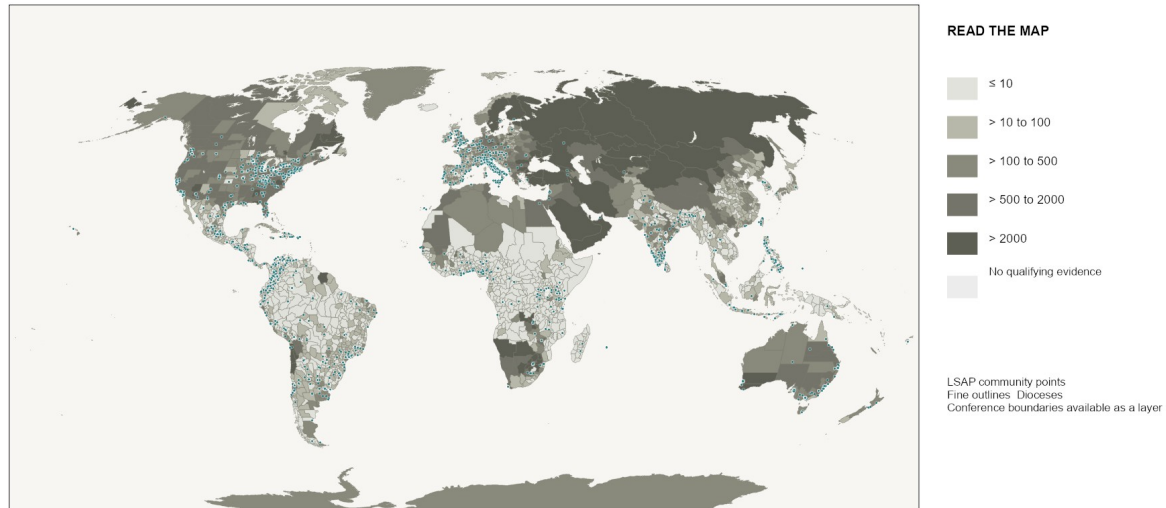
Mapped landfill footprints across dioceses

This report explains the source data, executed workflow and interpretation of mapped landfill footprints across dioceses. It documents the existing ArcGIS Pro output, including its coverage limits and retained ecclesiastical context. (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

Map P09 | Documentation date 6 October 2026 | Units Hectares | Inventory retrieved 6 October 2026

Mapped landfill footprints across dioceses

Unioned mapped landfill footprint · hectares within diocese



Footprints split at diocesan boundaries; overlapping footprint area counted once. Generic landfill tags do not establish pollution severity.

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LSAP POLLUTION ATLAS / COPIED ECCLESIASTICAL AND MEMBER FOUNDATIONS / P09

Figure P09 World overview exported from the completed ArcGIS Pro layout.

Footprints split at diocesan boundaries; overlapping footprint area counted once. Generic landfill tags do not establish pollution severity.

All analyses use copies of 2,874 dioceses and 2,739 LSAP member points. Copies of 122 episcopal conference features and 194 country features preserve ecclesiastical context. Stable DIO_ID links outputs to the copied boundaries. Original foundational data were not edited (LSAP Pollution Atlas, 2026, foundation_manifest.json).

Analysis workflow

The following numbered sequence describes the executed ArcPy and Python analyses as a typed ModelBuilder style workflow. Custom raster, station aggregation and geometry operations are script steps, not a claim that an ArcGIS ModelBuilder model file was run. Source data: (OpenStreetMap contributors, 2026; OpenStreetMap Wiki contributors, n.d.).

1. Use repaired working footprint copies and copied diocesan boundaries.
2. Intersect footprints and boundaries in equal-area EPSG:6933; union clipped footprints by diocese so overlapping mapped polygons count once.
3. Calculate unioned Footprint_ha; divide by full geodesic diocesan polygon area to calculate AreaPct. Retain site counts alongside area metrics.
4. Join to all copied dioceses; classify footprint hectares at 10, 100, 500 and 2,000. Expose AreaPct and count metrics in popups.
5. Keep original source, working footprints and derivative summaries; credit © OpenStreetMap contributors and ODbL.

Output fields and files

Field or band	Meaning
SiteCount	Sites uniquely assigned by greatest footprint overlap
IntersectSites	Distinct footprints touching the diocese
Footprint_ha	Unioned clipped footprint area in hectares
AreaPct	Footprint area as percent of full geodesic polygon area
Sites1000km2	Assigned sites per 1000 square kilometres
WasteStory	Partial inventory narrative

Analysis dataset: PollutionAnalysis.gdb/Dioceses_Landfills. Styles/P09.lyrx and Styles/P09.mapx preserve layer and map definitions. Map_Exports/P09_World_overview.png preserves the layout preview. The editable project is LSAP_Pollution.aprx.

Interpretation and data handling

Mapped footprint area is not contaminated area. A generic landfill tag does not distinguish controlled landfill from open dump. OSM tag definition.

The working inventory contains 51,748 unique sites after removing 2,529 duplicate records. Of these, 50,996 receive a primary diocese assignment and 752 remain unassigned. Raw regional JSON, OSM_Landfill_Source and Landfills_Working remain available. Site control status and pollution severity are unknown. The European mirror is older than most other regions; retrieval date does not imply a uniform data vintage (LSAP Pollution Atlas, 2026, waste_QA.json).

Cartographic classes are descriptive display intervals, not regulatory or ecological quality grades. The pollution ramp is #E1E2DB, #B7B8A9, #898A7C, #74746A and #5D5E54. Lower oxygen and shorter proximity distances use the dark end. Foundation member dots use teal #247C80; proximity maps color member points by distance.

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